

01_linear_regression

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[ ]: import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
import seaborn as sns
from IPython.display import display
```

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RANDOM_STATE = 42
sns.set_theme(style="whitegrid")
plt.rcParams.update({"figure.dpi": 120})

DATA_DIR = Path("../data")
if not DATA_DIR.exists():
    DATA_DIR = Path("Durga/Lab/MLDL/data")

pd.set_option("display.max_columns", 120)
```

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[2]: from sklearn.compose import ColumnTransformer
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
from sklearn.model_selection import train_test_split
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import OneHotEncoder, StandardScaler

df = pd.read_csv(DATA_DIR / "1000_companies.csv", encoding="utf-8-sig")
display(df.head())
display(df.describe(include="all"))
```

	R&D Spend	Administration	Marketing Spend	State	Profit
0	165349.20	136897.80	471784.10	New York	192261.83
1	162597.70	151377.59	443898.53	California	191792.06
2	153441.51	101145.55	407934.54	Florida	191050.39
3	144372.41	118671.85	383199.62	New York	182901.99
4	142107.34	91391.77	366168.42	Florida	166187.94

	R&D Spend	Administration	Marketing Spend	State \
count	1000.000000	1000.000000	1000.000000	1000
unique	NaN	NaN	NaN	3
top	NaN	NaN	NaN	California
freq	NaN	NaN	NaN	344

mean	81668.927200	122963.897612	226205.058419	NaN
std	46537.567891	12613.927535	91578.393542	NaN
min	0.000000	51283.140000	0.000000	NaN
25%	43084.500000	116640.684850	150969.584600	NaN
50%	79936.000000	122421.612150	224517.887350	NaN
75%	124565.500000	129139.118000	308189.808525	NaN
max	165349.200000	321652.140000	471784.100000	NaN

	Profit
count	1000.000000
unique	NaN
top	NaN
freq	NaN
mean	119546.164656
std	42888.633848
min	14681.400000
25%	85943.198543
50%	117641.466300
75%	155577.107425
max	476485.430000

```
[3]: target = "Profit"
X = df.drop(columns=[target])
y = df[target]

numeric_features = ["R&D Spend", "Administration", "Marketing Spend"]
categorical_features = ["State"]

preprocess = ColumnTransformer(
    transformers=[
        ("num", StandardScaler(), numeric_features),
        ("cat", OneHotEncoder(drop="first", handle_unknown="ignore"),
        categorical_features),
    ]
)

model = Pipeline(
    steps=[
        ("preprocess", preprocess),
        ("regressor", LinearRegression()),
    ]
)

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=RANDOM_STATE
)
model.fit(X_train, y_train)
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predictions = model.predict(X_test)

metrics = pd.DataFrame(
    {
        "metric": ["MAE", "RMSE", "R2"],
        "value": [
            mean_absolute_error(y_test, predictions),
            np.sqrt(mean_squared_error(y_test, predictions)),
            r2_score(y_test, predictions),
        ],
    }
)
display(metrics)

```

	metric	value
0	MAE	1404.437848
1	RMSE	5547.534845
2	R2	0.982528

```

[4]: feature_names = model.named_steps["preprocess"].get_feature_names_out()
coef_df = pd.DataFrame(
    {
        "feature": feature_names,
        "coefficient": model.named_steps["regressor"].coef_,
    }
).sort_values("coefficient", key=lambda s: s.abs(), ascending=False)
display(coef_df)

fig, axes = plt.subplots(1, 2, figsize=(12, 4))
axes[0].scatter(y_test, predictions, edgecolor="black", alpha=0.75)
lims = [min(y_test.min(), predictions.min()), max(y_test.max(), predictions.
    ↪max())]
axes[0].plot(lims, lims, color="black", linestyle="--")
axes[0].set_title("Actual vs Predicted Profit")
axes[0].set_xlabel("Actual")
axes[0].set_ylabel("Predicted")

residuals = y_test - predictions
axes[1].scatter(predictions, residuals, edgecolor="black", alpha=0.75)
axes[1].axhline(0, color="black", linestyle="--")
axes[1].set_title("Residuals")
axes[1].set_xlabel("Predicted")
axes[1].set_ylabel("Actual - Predicted")
plt.tight_layout()
plt.show()

```

	feature	coefficient
0	num__R&D Spend	24881.652357

1	num__Administration	13174.603836
2	num__Marketing Spend	7641.483136
3	cat__State_Florida	-874.491486
4	cat__State_New York	-97.133799

